Lesson plan · for the teacher

# Relationships

Data science · Level 2 · Lesson 3

**Time**
40 min

**Materials**
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

## Learning goals

- Draw and read a scatter plot.
- Read a correlation r.
- Ask what else could explain a pattern.

## Activity: A class scatter chart

1. Warm-up (5 min): Ask pupils to guess, then explain that we need both numbers for each pupil to find out.
2. Teach (8 min): Sketch rising, falling and cloud patterns with their r. Tell the ice cream story.
3. Plot (15 min): Pupils measure in pairs and add one dot each to axes on the board.
4. Discuss (7 min): Is the pattern strong? Then read two school claims and ask what else could explain each.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

## Discussion

- Why is height linked to arm span?
- What could explain sleep and marks going together?
- How would you test a cause fairly?

## Exit ticket

- Sketch a scatter plot with r close to -0.9.

**Try it at school tomorrow**
Pupils measure their height and arm span and add one dot each to a scatter chart on the board. Is the pattern strong? Then discuss a pattern that is not a cause.

# Relationships



**1** **More books at home goes with higher marks (r = 0.6). Does that prove books cause higher marks?**

◯ Only if r is 1 ◯ No: something else may explain both ◯ Yes

**2** **Match each pattern to its correlation.**

| | | |
|---|---|---|
| Dots close to a rising line | • • • • | About -0.9 |
| A shapeless cloud | • • • • | About 0.9 |
| Dots close to a falling line | • • • • | About 0 |

**3** **Finish the program so it prints the correlation r between study hours and marks, rounded to 2 decimal places.**

```
hours = [1, 2, 3, 4, 5]
marks = [8, 9, 11, 12, 15]
mx = sum(hours) / 5
my = sum(marks) / 5
sxy = 0
sxx = 0
syy = 0
for i in range(5):
    sxy = sxy + (hours[i] - mx) * (marks[i] - my)
print(sxy)
```

Write your program:

**4** **Sleep and marks go together. Which could also explain it? Choose all that apply.**

☐ The colour of the school uniform ☐ A calm home with help for homework

☐ Living close to school ☐ Good health

**5** **One unusual pupil can change the correlation a lot.**

◯ True ◯ False

✎ **Which school "fact" have you heard that might be a correlation, not a cause?**

Check your answers

# Relationships



**1** **No: something else may explain both**

Even a strong correlation is not proof of cause.

**2** **Dots close to a rising line — About 0.9**

**A shapeless cloud — About 0**

**Dots close to a falling line — About −0.9**

The sign shows the direction and the size shows how close to a line.

**3**

```
hours = [1, 2, 3, 4, 5]
marks = [8, 9, 11, 12, 15]
mx = sum(hours) / 5
my = sum(marks) / 5
sxy = 0
sxx = 0
syy = 0
for i in range(5):
    sxy = sxy + (hours[i] - mx) * (marks[i] - my)
    sxx = sxx + (hours[i] - mx) ** 2
    syy = syy + (marks[i] - my) ** 2
print(round(sxy / (sxx * syy) ** 0.5, 2))
```

$r = 17 \div \sqrt{(10 \times 30)} \approx 0.98$: a very strong positive relationship in these five pupils, which still doesn't prove that study alone causes the marks.

**4** **A calm home with help for homework; Good health; Living close to school**

Each of the first three could raise both sleep and marks.

**5** **True**

With few dots, one far-away point can pull r up or down, so always look at the chart.